

Algebra 1
Unit 2
Lesson 1 Homework

Name Answer Key
Date _____
Period _____

1. The product of powers law of exponents states to keep the then _____ the powers.

☐ add
☐ subtract
☐ multiply

☐ base
☐ coefficient and
☐ exponent

2. Jeffrey simplified the algebraic expression and made an error. Find and fix the mistake.

$$\begin{aligned} &(-2x^2y^3)^4(3.5x^0y) \\ &\quad (-2x^8y^{12})(3.5y) \\ &(-2 \cdot 3.5)(x^8)(y^{12} \cdot y) \\ &\quad -7.5x^8y^{13} \end{aligned}$$

$$\begin{aligned} &(-2x^2y^3)^4(3.5x^0y) \\ &\quad (16x^8y^{12})(3.5y) \\ &\quad 56x^8y^{13} \end{aligned}$$

3. Write a note to Jeffrey explaining the error he made in the previous question.

Apply the Power of a Product Law of Exponents to the numerical coefficients.

Determine an equivalent algebraic monomial expression for each expression using the laws of exponents. Show your work.

4. $(-4a^2b)(5ab^3)$

$$-20a^3b^4$$

5. $(\frac{-1}{2}x^3y^0)(6x^2y^3)(-3y)$

$$9x^5y^4$$

$$6. \quad (5.6kj^8)(-3.25k^9)(\frac{1}{4}k^7j^{12})$$

$$-\frac{91}{20}k^{17}j^{20} \quad OR \quad -4.55k^{17}j^{20}$$

$$7. \quad (-5mn^3p^5)^2$$

$$25m^2n^6p^{10}$$

$$8. \quad (4.1c^{-4}d)(-2.2c^{10}d^0)(-c^{-2}d^3)$$

$$9.02c^4d^4$$

$$9. \quad (\frac{5}{8}r^{-2})^0(-7.6r^4s^5)^{-2}$$

$$\frac{25}{1444r^8s^{10}} \quad OR \quad \frac{1}{57.76r^8s^{10}}$$

$$10. \quad (-2y^3z)^5(6y^{10}z^8)^2$$

$$-1152y^{35}z^{21}$$